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SEPM ASSIGNMENT - 3

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PS To understand DevOps: Principles, Practices and DevOps Engineer Role and Responsibilities

Ans DevOps is a combination of development and operations that focuses on improving collaboration, automation, and continuous delivery of Software.

It helps organizations:
Deliver software faster
Improve quality and reliability
Reduce Development failures

A) DevOps Principles
① Collaboration - Developers and operation teams work together with shared responsibility
② Automation - Automates repetitive tasks like testing, building and development
③ Continuous Integration (CI)
Code is integrated frequently into a shared repository. Automated testing ensures early bug detection
④ Continuous Delivery (CD)
Ensures faster and safer releases

(5) Continuous Monitoring - Tracks System performance and defects issues

(6) Infrastructure as code (IaC) Infrastructure is managed using code instead of manual setup Ensure consistency accross environments

Q Case study: DevOps Transformation at Netflix

Netflix is a global streaming service provider serving millions of users worldwide.

B) PS (Before DevOps)
Before adopting DevOps practices Netflix faced several issues:

- Slow software delivery
- Frequent failures during updates
- Poor teamwork between Dev & Ops
- Manual work causing errors
- Hard to scale systems

Root issue
Dev focuses on features, Ops focuses on stability

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DevOps Implementation
Netflix adopted DevOps to overcome these challenges by focusing on:

DevOps Principles
Collaboration, Automation, Continuous Feedback, Customer Centric Delivery (as discussed in A)

DevOps Practices Used
Continuous Integration (CI) - Developers frequently commit code
Automated builds and testing are triggered
- Continuous Delivery and Deployment (CD) - Faster release cycles (multiple times per day)
- Continuous Testing - Automated testing ensures bug detection early
- Infrastructure as Code (IaC)
Infrastructure managed using code
Ensures consistent ~~features~~ environments (dev, test, production)
- Microservice Architecture - Improves scalability and flexibility
- Continuous Monitoring - Quick detection and recovery from failures

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D) Role of DevOps Engineer
DevOps engineers play a crucial role
Responsibilities include :
- Automate building, testing and deployment pipelines
- Managing cloud infrastructure
- Ensuring system reliability and uptime
- Implementing security and compliance practices
- Facilitating collaboration

E) Results after DevOps adoption
- Faster deployments (many releases per day)
- Less downtime, quick recovery
- Better user experience
- High scalability
- Faster innovation and delivery of feature

Transforms traditional software development into fast, reliable and scalable systems

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f) Key learning
- DevOps is not just tools - It is a culture and mindset
- Automation is essential for speed and reliability
- Collaboration reduces conflicts and improves efficiency
- Continuous practices (CI/CD/testing/monitoring) ensure quality

g) Tools used (commonly used DevOps tools)
- CI/CD : Jenkins
- Containerization : Docker
- Orchestration : Kubernetes
- Monitoring : Prometheus, Grafana
- Cloud : AWS